

THE MARINE FOOD WEB

Nearly all life in the ocean depends on phytoplankton, which turn simple chemicals into complex substances that animals can use as food. The phytoplankton are eaten by microscopic animals, which are in turn eaten by slightly larger animals. They all drift and swim in the oceans as zooplankton. These plankton swarms are harvested by larger creatures like fish, which fall prey to hunters such as sharks.

ZOOPLANKTON



RADIOLARIANS AND FORAMS

These microscopic protozoans multiply fast among blooms of phytoplankton such as diatoms and dinoflagellates, which they catch by seizing them with threads of body tissue that they extend through holes in their shells. Radiolarians like these have shells of glassy silica. By contrast, forams have lime-rich shells that resemble tiny snails.



COPEPODS

Copepods are miniature crustaceans (shrimplike animals) that trap phytoplankton by straining water through their feathery legs. They form dense swarms in food-rich waters, providing prey for shoaling fish. At night they swim up toward the surface, where phytoplankton are most abundant, but they retreat to the darker, safer twilight zone at dawn.



KRILL

In the cold Southern Ocean around Antarctica, copepods are replaced by krill. These much bigger crustaceans feed in the same way as copepods, but they form larger swarms that can turn the ocean water red. They are the main food of most Antarctic fish, seabirds, penguins, crabeater seals, and whales, and without them the Antarctic ecosystem would collapse.



EGGS AND LARVAE

The drifting community of plankton forms a nursery for the eggs and larvae of many marine animals, including fish, mollusks, and crustaceans. When the eggs hatch, the larvae feed on plankton until they turn into small adults. Many, like these crab larvae, will then begin a completely different way of life, feeding on the seabed or even clamped onto rocks.



▲ DRIFTING JELLIES

All kinds of strange jellylike animals hunt among the plankton.

Although many can swim, they mainly drift in the currents with their prey. They include jellyfish like this one, which is using its poisonous tentacles to trap copepods, as well as iridescent comb jellies and organisms called salps that look like long chains of plastic bags.



▲ HUNGRY SHOALS

Many small fish such as anchovies and herring feed by swimming through plankton swarms with their mouths open, so that the water flows through their gills. The gridlike gill rakers protecting their gills act as sieves, trapping plankton. These fish swim in dense shoals of thousands or millions, each shoal forming a superorganism that behaves like one huge animal.